

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

I _Bharath Sankar (192425046)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) Clearly interpret the scenario and explain the cause of low contrast.

A low-contrast image looks dull because the object and background have similar gray intensity values. This makes it difficult to distinguish objects.

(b) Identify all key issues affecting image visibility.

- Poor visibility
- Low brightness difference
- Difficult object detection
- Loss of image details

(c) Apply an appropriate enhancement technique to address the problem.

Histogram Equalization

How it works:

- Spreads pixel intensity values.
- Increases contrast.
- Makes objects more visible.

(d) Justify your choice with logical reasoning.

Histogram Equalization improves overall contrast without changing the image content. It is simple and effective for low-contrast grayscale images.

(e) Present your explanation in a clear and structured manner.

Histogram Equalization enhances visibility and makes the image clearer for further processing.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpret the scenario and explain the impact of uneven illumination.

The image contains bright and dark regions due to non-uniform lighting, making analysis difficult.

(b) Identify key factors causing intensity variation.

- Uneven brightness
- Loss of details in dark areas
- Overexposed bright regions
- Poor segmentation

(c) Apply a suitable enhancement method to normalize illumination.

Adaptive Histogram Equalization (CLAHE)

How it works:

- Enhances small image regions separately.
- Improves local contrast.
- Prevents over-enhancement.

(d) Justify your selected approach with reasoning.

CLAHE corrects uneven lighting while preserving image details, making it better than normal Histogram Equalization.

(e) Provide a clear and well-structured explanation.

CLAHE produces a uniformly illuminated image with improved visibility.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise.

Salt-and-pepper noise appears as random black and white pixels, reducing image readability.

(b) Identify key issues impacting image quality.

- Random black and white dots
- Reduced readability
- Distorted edges
- Poor image quality

(c) Apply an appropriate filtering technique.

Median Filter

How it works:

- Replaces each pixel with the median value of neighboring pixels.
- Removes impulse noise.

(d) Justify why this method is more suitable than alternatives.

Median Filter removes salt-and-pepper noise while preserving edges better than the Mean Filter.

(e) Present your response clearly and logically.

Median filtering effectively restores image quality and improves readability.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpret the scenario and explain why edges are blurred.

Excessive smoothing removes not only noise but also important edges, making the image blurry.

(b) Identify key issues related to smoothing and edge loss.

- Edge information lost
- Reduced sharpness
- Poor object boundaries
- Less accurate feature extraction

(c) Apply an appropriate technique to restore edges.

Unsharp Masking (Image Sharpening)

How it works:

- Enhances edge details.
- Increases sharpness.
- Restores clear boundaries.

(d) Justify your decision with clear reasoning.

Unsharp Masking sharpens important edges without significantly increasing noise.

(e) Structure your explanation clearly.

Image sharpening restores edge clarity and improves object detection.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpret the scenario and describe high-frequency noise.

The satellite image contains fine-grained high-frequency noise that hides important features.

(b) Identify the key issues affecting image clarity.

- Fine random noise
- Reduced feature visibility
- Poor image clarity
- Difficult image interpretation

(c) Apply an appropriate frequency domain filtering method.

Low-Pass Filter (Gaussian Low-Pass Filter)

How it works:

- Removes high-frequency noise.
- Preserves low-frequency image information.
- Produces a smoother image.

(d) Justify the choice of filter with reasoning.

A Gaussian Low-Pass Filter effectively reduces high-frequency noise while maintaining the overall image structure.

(e) Present your answer in a clear and organized manner.

Low-pass filtering improves satellite image quality and enhances feature visibility.